

Assessment (Week 3)

Description

Module/Week3 – Assignment 2

Topic: Structured Query Language; Join Operations; Beyond Simple SELECT Queries

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Course Title: INFO 531 - Data Warehousing and Analytics in the Cloud

Term name and year: Fall 2023

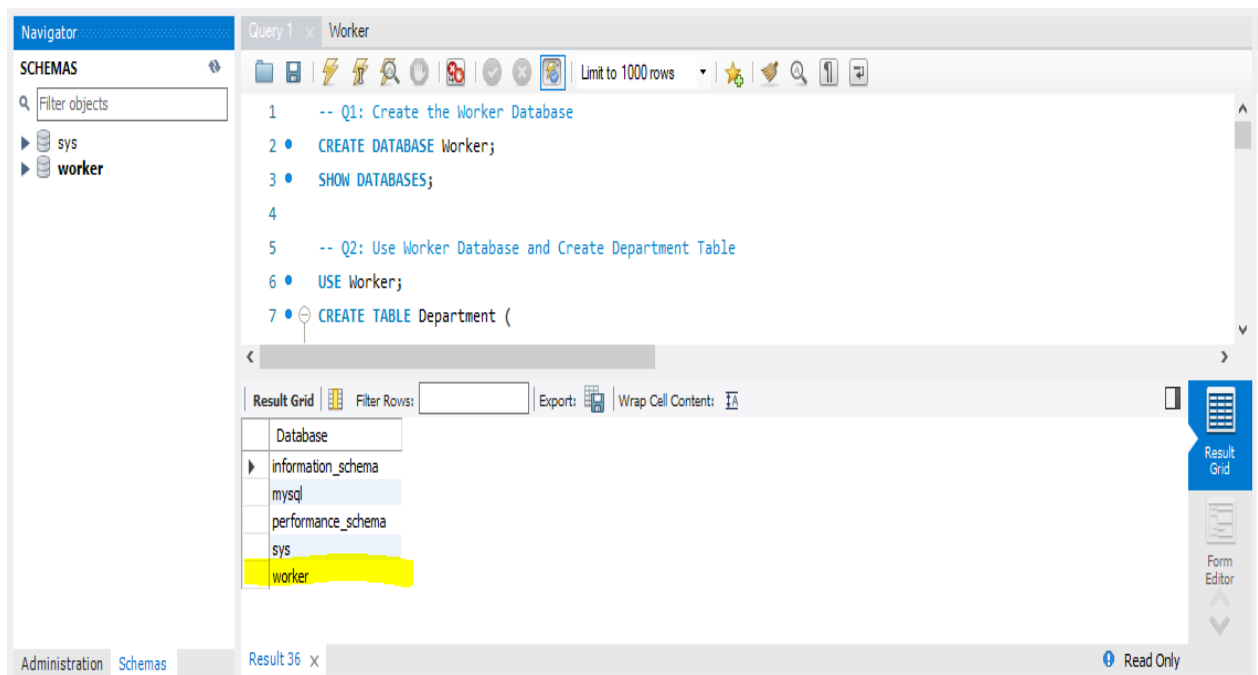
Submission Week: , Week 3- Assignment 2

Instructor's Name: Dr.Nayem Rahman

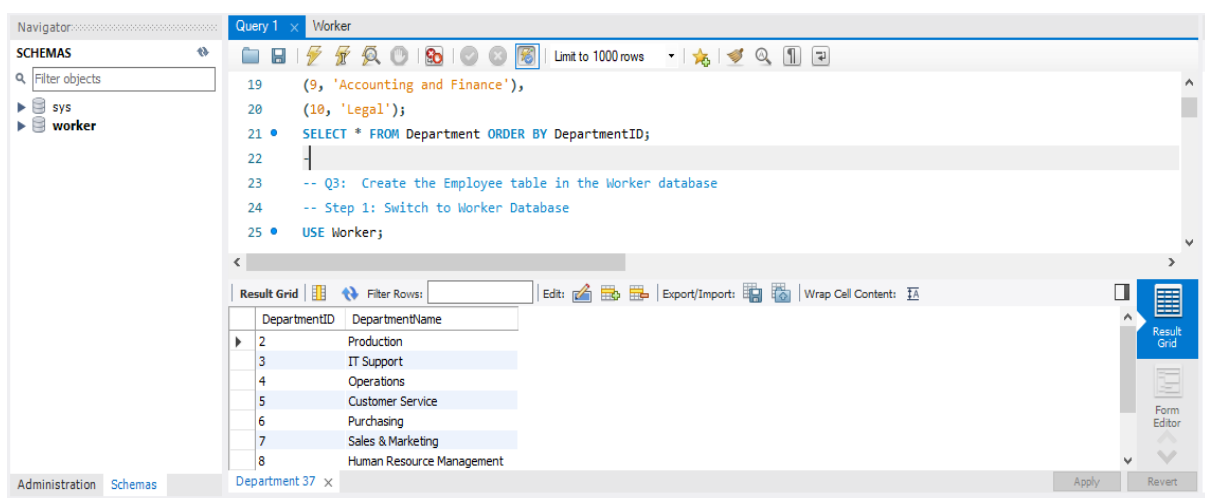
Date of Submission:31-07-2024

Q1. Show the screenshot of a successful installation of MySQL Software and MySQL Workbench with the latest version on your machine. Show the screenshot of the database “Worker” created.





Q2. . Create the Department table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Department table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



Q3. Create the Employee table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type and length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented (not in a graphical view). (c)

Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Employee table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).

The screenshot shows the SQL Developer interface. The query window contains the following SQL statements:

```

57 (4, 'Multi-Function Printers'),
58 (5, 'Projector or a Big Screen TV'),
59 (6, 'Servers'),
60 (7, 'Internet Modem'),
61 (8, 'Cell Phone');
62 • SELECT * FROM Equipment ORDER BY EquipmentID;
63 -- Q5: Create the EmployeeEquipment table in the Worker database

```

The result grid displays the output of the SELECT statement, showing employee data ordered by EmployeeID:

EmployeeID	DepartmentID	FirstName	LastName	Address	PhoneNumber	HireDate
1	2	Andy	Wong	345 South Street	(603) 555-6880	2001-01-15
2	1	John	Wilson	John Wilson	(518) 555-6690	2017-03-19
3	3	Vivek	Pandey	15 Mineral Drive	(603) 555-4420	2003-11-15
4	7	Nola	Davis	15 Long Ave	(478) 555-8822	2016-03-23
5	8	Kathy	Cooper	15 Hatter Drive	(212) 555-9630	2011-11-18
6	9	Tom	Harper	64 Highland Street	(212) 555-7755	2010-04-11
7	10	NULL	NULL	NULL	NULL	NULL

Q4. Create the Equipment table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Equipment table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).

The screenshot shows the SQL Developer interface. The query window contains the following SQL statements:

```

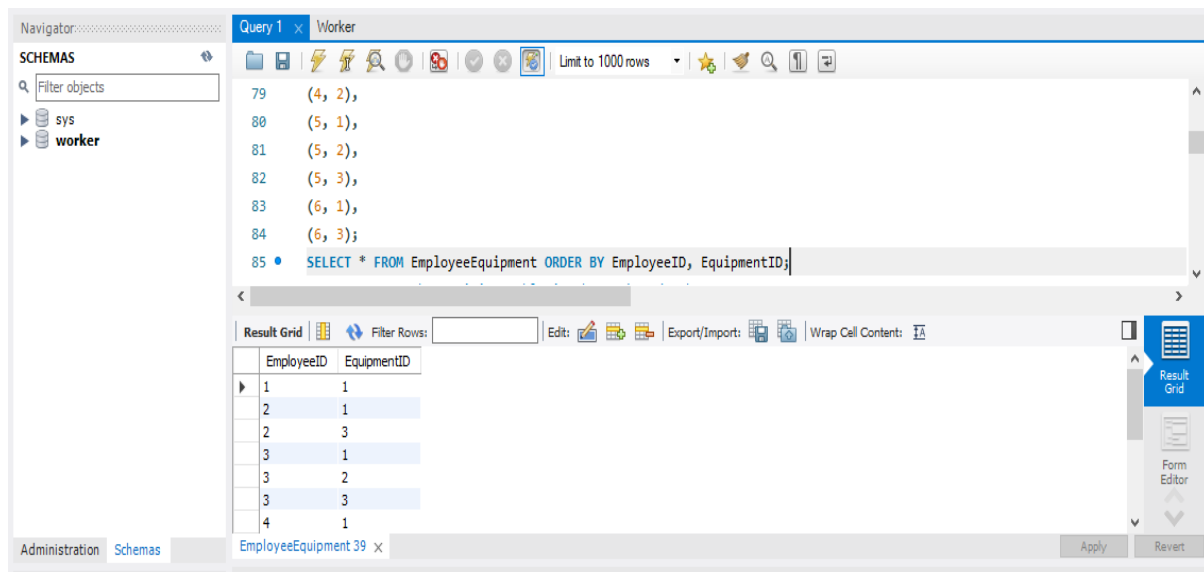
61 (8, 'Cell Phone', 10000);
62 • SELECT * FROM Equipment ORDER BY EquipmentID;
63 -- Q5: Create the EmployeeEquipment table in the Worker database
64 • CREATE TABLE EmployeeEquipment (
65     EmployeeID INTEGER NOT NULL,
66     EquipmentID INTEGER,
67     PRIMARY KEY (EmployeeID, EquipmentID),

```

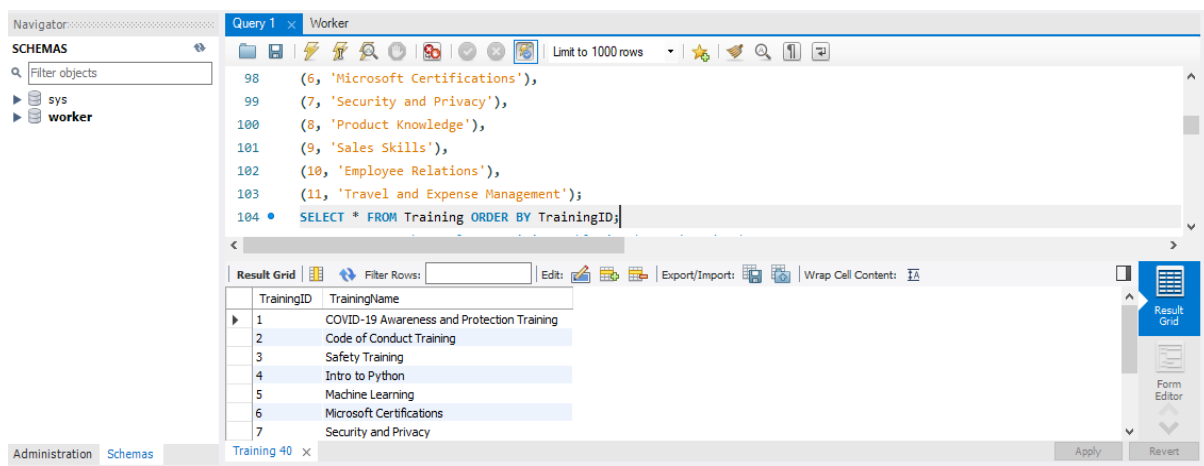
The result grid displays the output of the SELECT statement, showing equipment data ordered by EquipmentID:

EquipmentID	EquipmentName	EquipmentCostAmount
1	Notebook Computers	300.00
2	Headsets	400.00
3	Computer Monitor	500.00
4	Multi-Function Printers	600.00
5	Projector or a Big Screen TV	700.00
6	Servers	800.00
7	Internet Modem	900.00

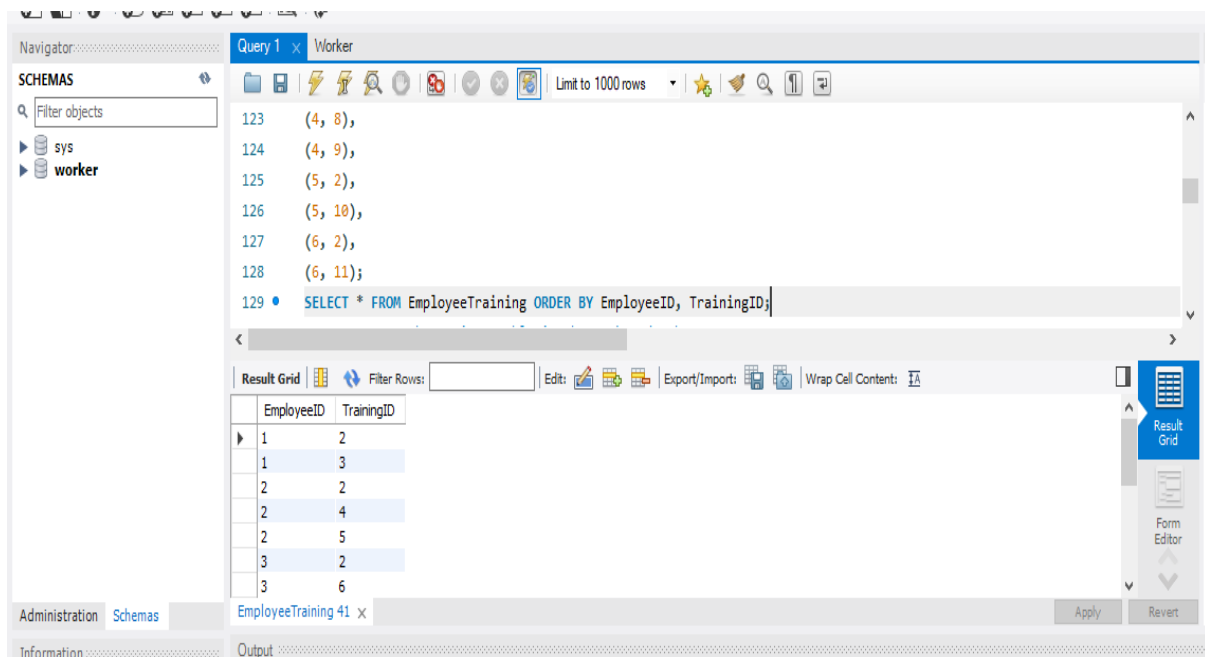
Q5. Create the EmployeeEquipment table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the EmployeeEquipment table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



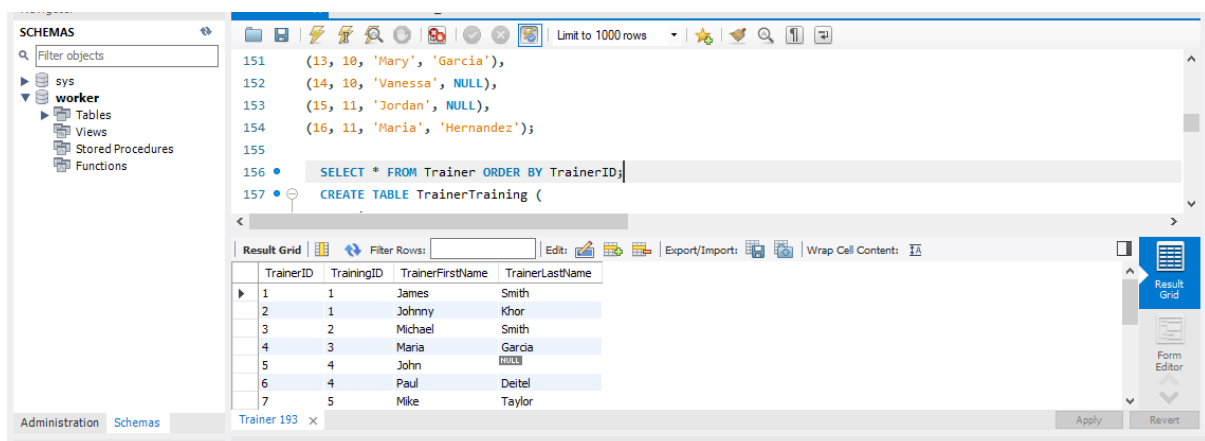
Q6. Create the Training table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Training table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



Q7. Create the EmployeeTraining table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the EmployeeTraining table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



Q8. Create the Trainer table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Trainer table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



Q9. Retrieve the data from the Trainer table by using the SELECT * statement with filter, WHERE TrainerLastName IS NULL. Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).

The screenshot shows the SQL Developer interface with a query window titled 'new worker'. The query is as follows:

```

181 • SELECT * FROM TrainerTraining ORDER BY TrainerID, TrainingID;
182
183 -- Q9: Retrieve the data from the Trainer table
184 • SELECT *
185 FROM Trainer
186 WHERE TrainerLastName IS NULL
187 ORDER BY TrainerID;

```

The result grid below the query shows the following data:

TrainerID	TrainingID	TrainerFirstName	TrainerLastName
5	4	John	NULL
14	10	Vanessa	NULL
15	11	Jordan	NULL
NULL	NULL	NULL	NULL

Q10. By using the SHOW tables statements, show the list of tables you have created in the Worker database. Show the screenshot of the execution of the above statements and results. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).

The screenshot shows the SQL Developer interface with a query window titled 'Query 1'. The query is as follows:

```

198
199
200 -- Q10: By using the SHOW tables statements, show the list of tables you have created in the Worker database
201 • SHOW TABLES;
202 -- Q11: Write a single-row subquery to display EmployeeID, FirstName, LastName, and HireDate of employees hired after
203 • SELECT HireDate
204 FROM Employee

```

The result grid below the query shows the following data:

Tables_in_worker
department
employee
employeeequipment
employeeequipment
equipment
trainer
trainertraining

Q11. Write a single-row subquery to display EmployeeID, FirstName, LastName, and HireDate of employees hired after employee Vivek Pandey. Sort the results by EmployeeID.

Navigator: Query 1 x Worker

SCHEMAS

Filter objects

sys

worker

```

208 FROM Employee
209 WHERE HireDate > (
210     SELECT HireDate
211     FROM Employee
212     WHERE FirstName = 'Vivek' AND LastName = 'Pandey'
213 )
214 ORDER BY EmployeeID;

```

Result Grid

EmployeeID	FirstName	LastName	HireDate
2	John	Wilson	2017-03-19
4	Nola	Davis	2016-03-23
5	Kathy	Cooper	2011-11-18
6	Tom	Harper	2010-04-11
NULL	NULL	NULL	NULL

Administration Schemas Employee 46 x Apply Revert

Q12. Write a query to display FirstName, LastName, and TrainingName for employee Tom Harper. Sort the results by TrainingName. Make sure you show the print screen of the complete set of the rows, and columns as specified.

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 x Worker

SCHEMAS

Filter objects

sys

worker

```

215 -- Q12:Write a query to display FirstName, LastName, and TrainingName for employee Tom Harper. Sort the results by Tra
216 SELECT e.FirstName, e.LastName, t.TrainingName
217 FROM Employee e
218 JOIN EmployeeTraining et ON e.EmployeeID = et.EmployeeID
219 JOIN Training t ON et.TrainingID = t.TrainingID
220 WHERE e.FirstName = 'Tom' AND e.LastName = 'Harper'
221 ORDER BY t.TrainingName;

```

Result Grid

FirstName	LastName	TrainingName
Tom	Harper	Code of Conduct Training
Tom	Harper	Travel and Expense Management

Administration Schemas Result 47 x Read Only

Q13. Write a query to display the complete list of Trainings, and trainers (first and last name) available for each training. Sort the output by TrainingName and Trainers' first and last name. Make sure you show the print screen of the complete set of the rows, and columns as specified.

Navigator: SCHEMAS
Filter objects
sys
worker

Query 1 Worker
Limit to 1000 rows

```

230 TrainerTraining tt ON t.TrainingID = tt.TrainingID
231 JOIN
232 Trainer tr ON tt.TrainerID = tr.TrainerID
233 ORDER BY
234 t.TrainingName,
235 tr.TrainerFirstName,
236 tr.TrainerLastName;

```

Result Grid Filter Rows: Export: Wrap Cell Content: 12

TrainingName	TrainerFirstName	TrainerLastName
Code of Conduct Training	Michael	Smith
COVID-19 Awareness and Protection Training	James	Smith
COVID-19 Awareness and Protection Training	Johnny	Khor
Employee Relations	Mary	Garcia
Employee Relations	Vanessa	NULL
Intro to Python	John	NULL
Intro to Python	Paul	Deitel

Administration Schemas Result 48 x Read Only

Q14. Write a multiple-row subquery to display EmployeeID, FirstName, LastName, and HireDate of employees who work for the following departments: Accounting and Finance, IT Support, and Production. Sort the results by EmployeeID. Make sure you show the print screen of the complete set of the rows, and columns as specified.

Navigator: SCHEMAS
Filter objects
sys
worker

Query 1 Worker
Limit to 1000 rows

```

247 e.DepartmentID IN (
248     SELECT d.DepartmentID
249     FROM Department d
250     WHERE d.DepartmentName IN ('Accounting and Finance', 'IT Support', 'Production')
251 )
252 ORDER BY
253 e.EmployeeID;

```

Result Grid Filter Rows: Edit: Export/Imports: Wrap Cell Content: 12

EmployeeID	FirstName	LastName	HireDate
1	Andy	Wong	2001-01-15
3	Vivek	Pandey	2003-11-15
6	Tom	Harper	2010-04-11
NULL	NULL	NULL	NULL

Administration Schemas Employee 49 x Apply Revert

Q15. Write a query to display the EmployeeID, FirstName, LastName, EquipmentName, and EquipmentCostAmount for one of the employees. Sort the results by EmployeeID. Make sure you show the print screen of the complete set of the rows, and columns as specified.

Query Editor: new_schema - Schema

```

250 JOIN
251   Equipment eq ON ee.EquipmentID = eq.EquipmentID
252 ORDER BY
253   e.EmployeeID;
254 -- Q16: Write a query to display the TrainingID, TrainingName, TrainerID, TrainerFirstName, and TrainerLastName wi
255 SELECT
256   t.TrainingID,

```

Result Grid (240 rows):

EmployeeID	FirstName	LastName	EquipmentName	EquipmentCostAmount
1	Andy	Wong	Notebook Computers	300.00
2	John	Wilson	Notebook Computers	300.00
2	John	Wilson	Computer Monitor	500.00
3	Vivek	Pandey	Notebook Computers	300.00
3	Vivek	Pandey	Headsets	400.00
3	Vivek	Pandey	Computer Monitor	500.00
4	Nola	Davis	Notebook Computers	300.00

Q16. Write a query to display the TrainingID, TrainingName, TrainerID, TrainerFirstName, and TrainerLastName with the trainers who did not provide their last name. Sort the results by TrainingID and TrainerID. Make sure you show the print screen of the complete set of the rows, and columns as specified.

Query Editor: Worker

```

280 JOIN
281   Trainer tr ON tt.TrainerID = tr.TrainerID
282 WHERE
283   tr.TrainerLastName IS NULL
284 ORDER BY
285   t.TrainingID,
286   tr.TrainerID;

```

Result Grid (52 rows):

TrainingID	TrainingName	TrainerID	TrainerFirstName	TrainerLastName
4	Intro to Python	5	John	NULL
10	Employee Relations	14	Vanessa	NULL
11	Travel and Expense Management	15	Jordan	NULL

Q17. Write a query to display the distinct list of equipments used by the current employees. Sort the output by EquipmentName. Make sure you show the print screen of the complete set of the rows, and columns as specified.

Navigator: SCHEMAS
Filter objects
sys
worker

Query 1 x Worker

Limit to 1000 rows

```

292 Employee e
293 JOIN
294 EmployeeEquipment ee ON e.EmployeeID = ee.EmployeeID
295 JOIN
296 Equipment eq ON ee.EquipmentID = eq.EquipmentID
297 ORDER BY
298 eq.EquipmentName;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

EquipmentName
Computer Monitor
Headsets
Notebook Computers

Administration Schemas Result 53 x Read Only

Q18. Write a query to display the FirstName, LastName, TrainingName, and trainer(s) (with first and last name in two separate columns) for one of the employees. Sort the results by TrainingName and TrainerFirstName. Make sure you show the print screen of the complete set of the rows, and columns as specified.

Navigator: SCHEMAS
Filter objects
sys
worker

Query 1 x Worker

Limit to 1000 rows

```

315 Trainer tr ON tt.TrainerID = tr.TrainerID
316 WHERE
317 e.EmployeeID = 1 -- Replace 1 with the desired EmployeeID
318 ORDER BY
319 t.TrainingName,
320 tr.TrainerFirstName;
321

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

EmployeeFirstName	EmployeeLastName	TrainingName	TrainerFirstName	TrainerLastName
Andy	Wong	Code of Conduct Training	Michael	Smith
Andy	Wong	Safety Training	Maria	Garcia

Administration Schemas Result 54 x Read Only

Q19. Write a query to display the EmployeeID, FirstName, LastName, DepartmentID, DepartmentName, EquipmentID, EquipmentName for all employees. Sort the results by EmployeeID, DepartmentID, and EquipmentID. Make sure you show the print screen of the complete set of the rows, and columns as specified.

Navigator: SCHEMAS
Filter objects
sys
worker

Query 1 x Worker

```

336 EmployeeEquipment ee ON e.EmployeeID = ee.EmployeeID
337 JOIN
338 Equipment eq ON ee.EquipmentID = eq.EquipmentID
339 ORDER BY
340 e.EmployeeID,
341 e.DepartmentID,
342 ee.EquipmentID;

```

Result Grid

	EmployeeID	FirstName	LastName	DepartmentID	DepartmentName	EquipmentID	EquipmentName
1	Andy	Wong	2	Production	1	Notebook Computers	
3	Vivek	Pandey	3	IT Support	1	Notebook Computers	
3	Vivek	Pandey	3	IT Support	2	Headsets	
3	Vivek	Pandey	3	IT Support	3	Computer Monitor	
4	Nola	Davis	7	Sales & Marketing	1	Notebook Computers	
4	Nola	Davis	7	Sales & Marketing	2	Headsets	
5	Kathy	Cooper	8	Human Resource Management	1	Notebook Computers	

Result 55 x Read Only

Q20. Write a query to display the EmployeeID, FirstName, LastName, DepartmentID, DepartmentName, TrainingID, TrainingName for all employees. Sort the results by EmployeeID, DepartmentID, and TrainingID. Make sure you show the print screen of the complete set of the rows, and columns as specified.

Navigator: SCHEMAS
Filter objects
sys
worker

Query 1 x Worker

```

357 EmployeeTraining et ON e.EmployeeID = et.EmployeeID
358 JOIN
359 Training t ON et.TrainingID = t.TrainingID
360 ORDER BY
361 e.EmployeeID,
362 e.DepartmentID,
363 et.TrainingID;

```

Result Grid

	EmployeeID	FirstName	LastName	DepartmentID	DepartmentName	TrainingID	TrainingName
1	Andy	Wong	2	Production	2	Code of Conduct Training	
1	Andy	Wong	2	Production	3	Safety Training	
3	Vivek	Pandey	3	IT Support	2	Code of Conduct Training	
3	Vivek	Pandey	3	IT Support	6	Microsoft Certifications	
3	Vivek	Pandey	3	IT Support	7	Security and Privacy	
4	Nola	Davis	7	Sales & Marketing	2	Code of Conduct Training	
4	Nola	Davis	7	Sales & Marketing	8	Product Knowledge	

Result 56 x Read Only